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Department:

Software Engineering

Group:

Computer Engineering Technology

Semester:

4th Semester

Subject:

Database Applications

Lab Task:

SQL Server – DDL Commands

Submitted To:

Engr. Asma Javid

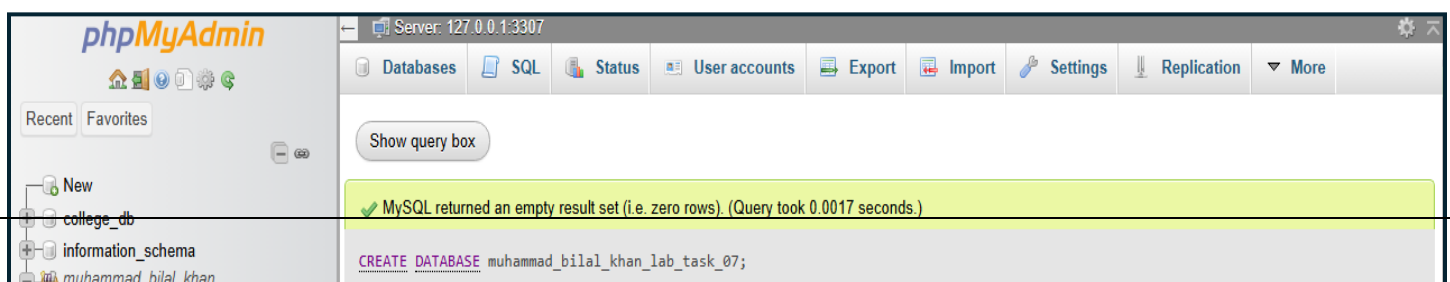
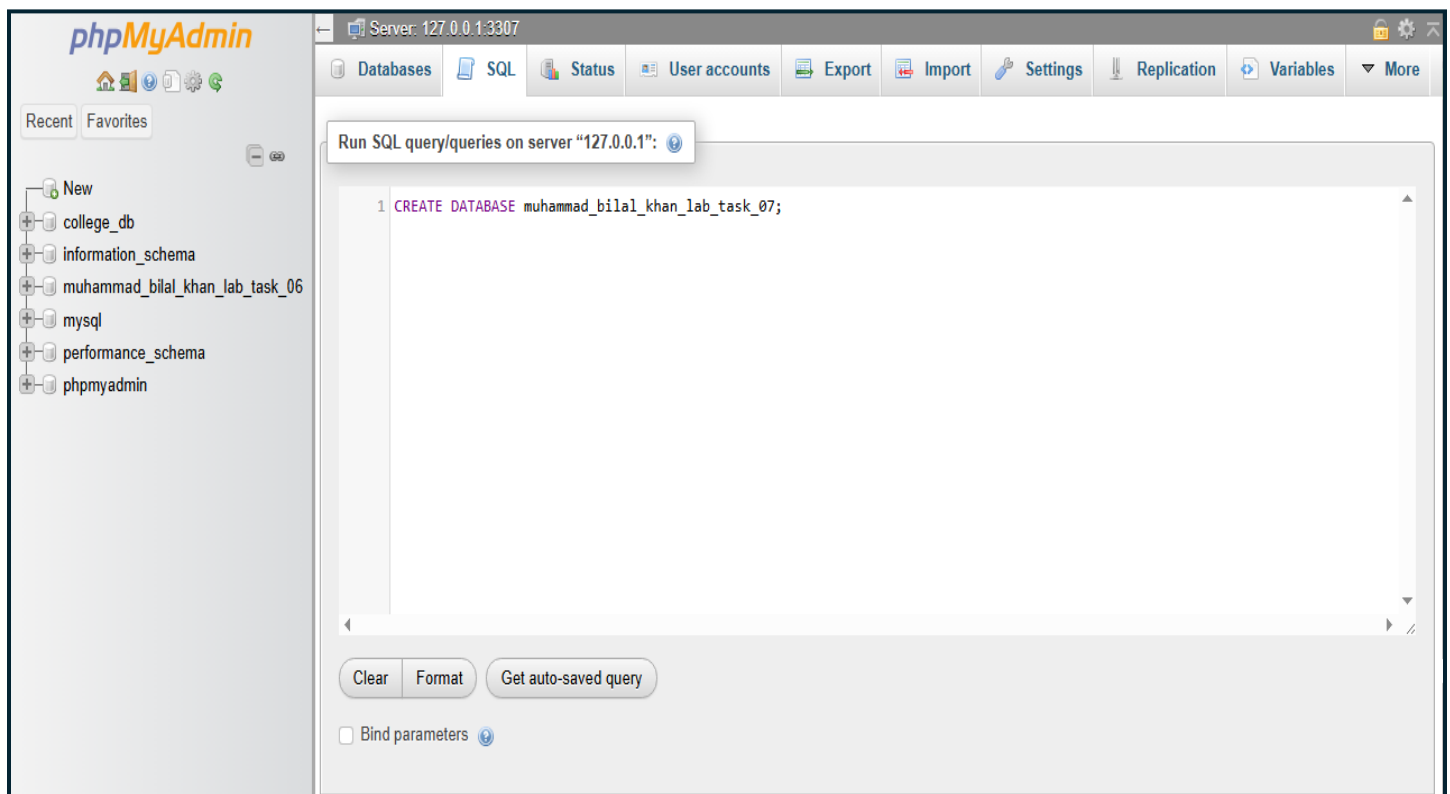
Lab Task 7

Section 1:

Creating a Database and Table Using SQL DDL Commands

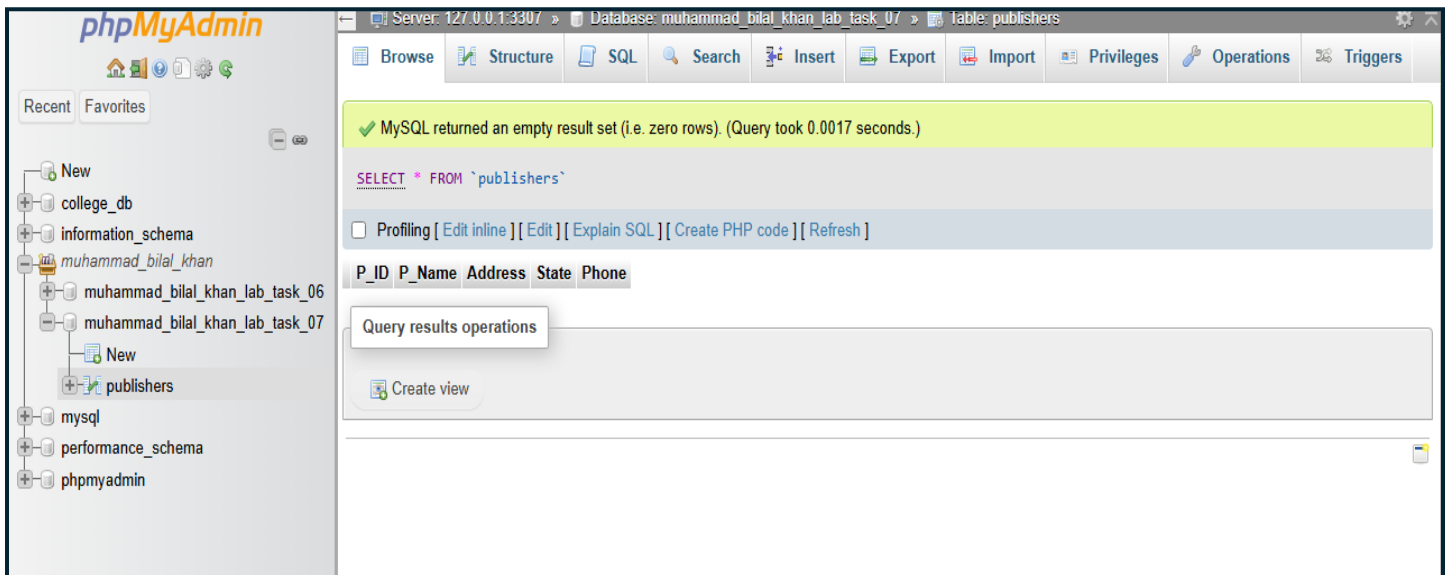
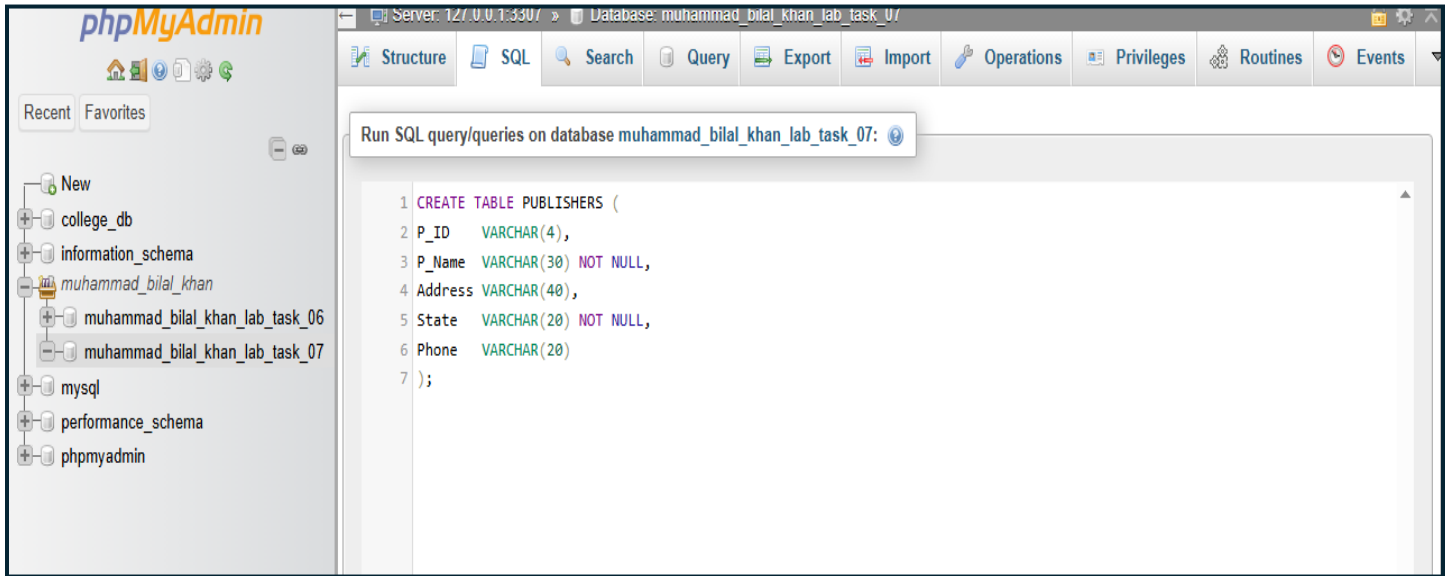
1. Create a new database named as your full name in all lowercase with underscores, followed by _lab_task_07.

Example: kainat_pervaiz_lab_task_07



2. In this database, use SQL DDL commands to create a table named PUBLISHERS with the following attributes:

Attribute	Data Type	Field Size
P_ID	VARCHAR	4
P_Name	VARCHAR	30
Address	VARCHAR	40
State	VARCHAR	20
Phone	VARCHAR	20

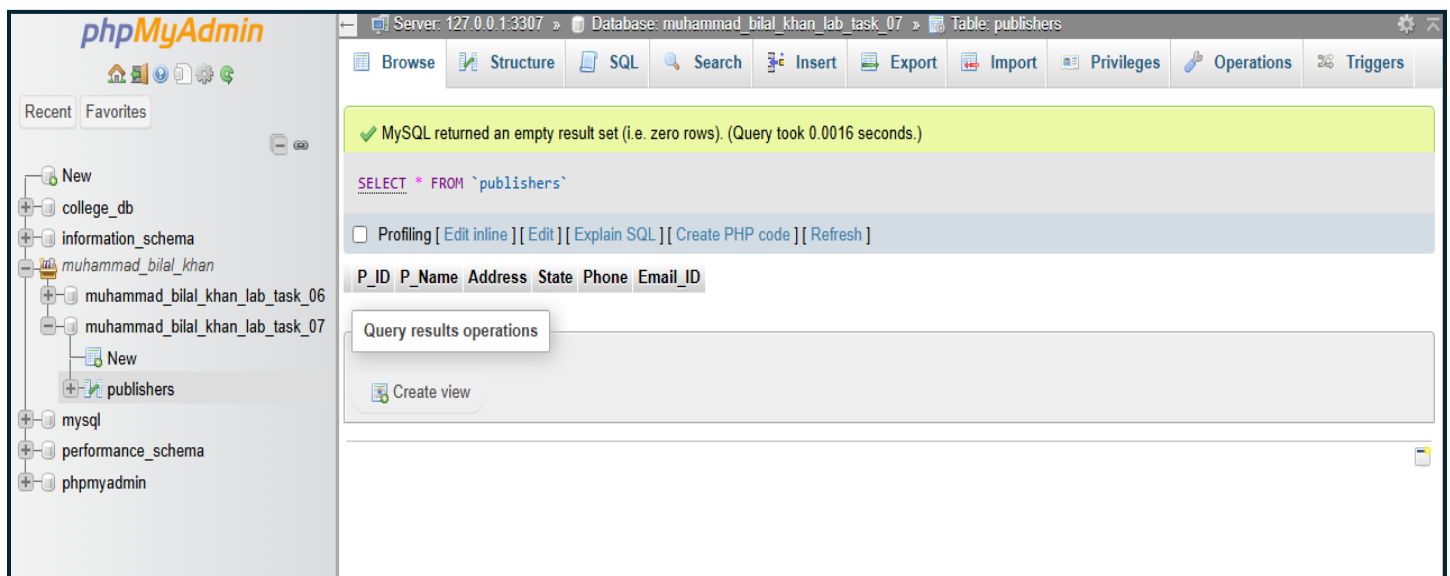
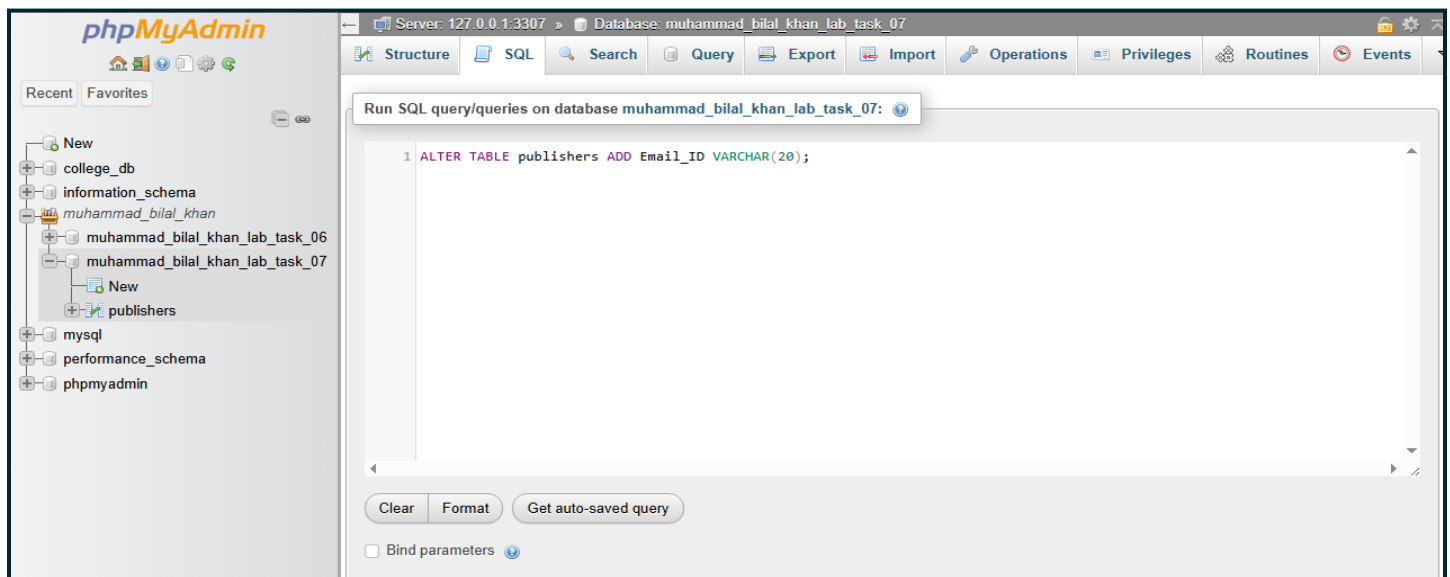


Section 02:

Modifying Table Structure Using SQL DDL Commands

Use SQL DDL commands to make the following changes to the PUBLISHERS table:

1. Add a new column Email_ID of type VARCHAR(20).



2. Change the data type of P_ID from VARCHAR to INT.

phpMyAdmin

Recent Favorites

- New
- college_db
- information_schema
- muhammad_bilal_khan
 - muhammad_bilal_khan_lab_task_06
 - muhammad_bilal_khan_lab_task_07
 - New
 - publishers
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_07 » Table: publishers

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on table muhammad_bilal_khan_lab_task_07.publishers:

```
1 ALTER TABLE publishers MODIFY P_ID INT;
```

P_ID
P_Name
Address
State
Phone
Email_ID

SELECT * SELECT INSERT UPDATE DELETE Clear Format Get auto-saved query

☐ Bind parameters

phpMyAdmin

Recent Favorites

- New
- college_db
- information_schema
- muhammad_bilal_khan
 - muhammad_bilal_khan_lab_task_06
 - muhammad_bilal_khan_lab_task_07
 - New
 - publishers
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_07 » Table: publishers

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Table structure Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	P_ID	int(11)			Yes	NULL			Change Drop More
☐ 2	P_Name	varchar(30)	utf8mb4_general_ci		No	None			Change Drop More
☐ 3	Address	varchar(40)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 4	State	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
☐ 5	Phone	varchar(20)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 6	Email_ID	varchar(20)	utf8mb4_general_ci		Yes	NULL			Change Drop More

☐ Check all With selected: Browse Change Drop Primary Unique Index Spatial

Fulltext

3. Change the data type of Phone from VARCHAR to INT.

phpMyAdmin

Recent Favorites

- New
- college_db
- information_schema
- muhammad_bilal_khan
 - muhammad_bilal_khan_lab_task_06
 - muhammad_bilal_khan_lab_task_07
 - New
 - publishers
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_07 » Table: publishers

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on table muhammad_bilal_khan_lab_task_07.publishers:

```
1 ALTER TABLE publishers MODIFY Phone INT;
```

P_ID
P_Name
Address
State
Phone
Email_ID

SELECT * SELECT INSERT UPDATE DELETE Clear Format Get auto-saved query

☐ Bind parameters

phpMyAdmin

Recent Favorites

- New
- college_db
- information_schema
- muhammad_bilal_khan
 - muhammad_bilal_khan_lab_task_06
 - muhammad_bilal_khan_lab_task_07
 - New
 - publishers
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_07 » Table: publishers

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Table structure Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 P_ID	int(11)			Yes	NULL			Change Drop More
☐	2 P_Name	varchar(30)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 Address	varchar(40)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	4 State	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
☐	5 Phone	int(11)			Yes	NULL			Change Drop More
☐	6 Email_ID	varchar(20)	utf8mb4_general_ci		Yes	NULL			Change Drop More

☐ Check all
 With selected: Browse Change Drop Primary Unique Index Spatial

Fulltext

Print Propose table structure Move columns Normalize

Add 1 column(s) after Email_ID Go

Section 03:

Renaming and Deleting Table Using SQL DDL Commands

1. Rename the table PUBLISHERS to PUBLISHER using an SQL command.

phpMyAdmin

Recent Favorites

- New
- college_db
- information_schema
- muhammad_bilal_khan
 - muhammad_bilal_khan_lab_task_06
 - muhammad_bilal_khan_lab_task_07
 - New
 - publishers
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_07 » Table: publishers

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on table muhammad_bilal_khan_lab_task_07.publishers:

```
1 ALTER TABLE publishers RENAME TO publisher;
```

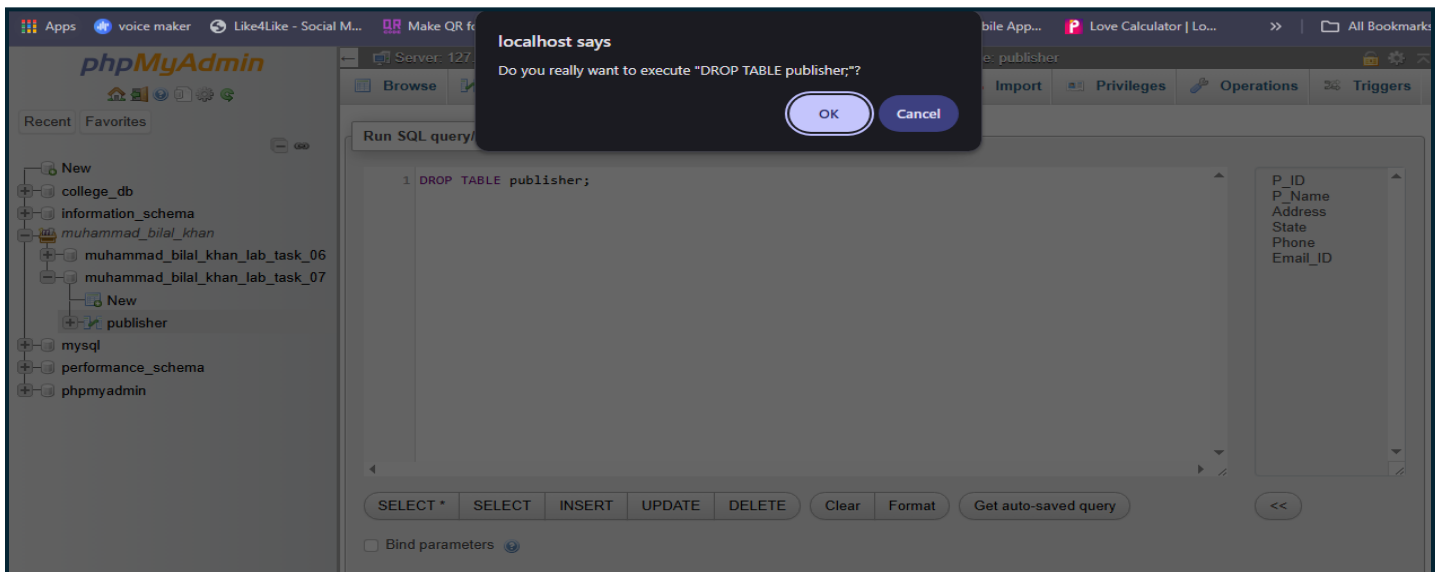
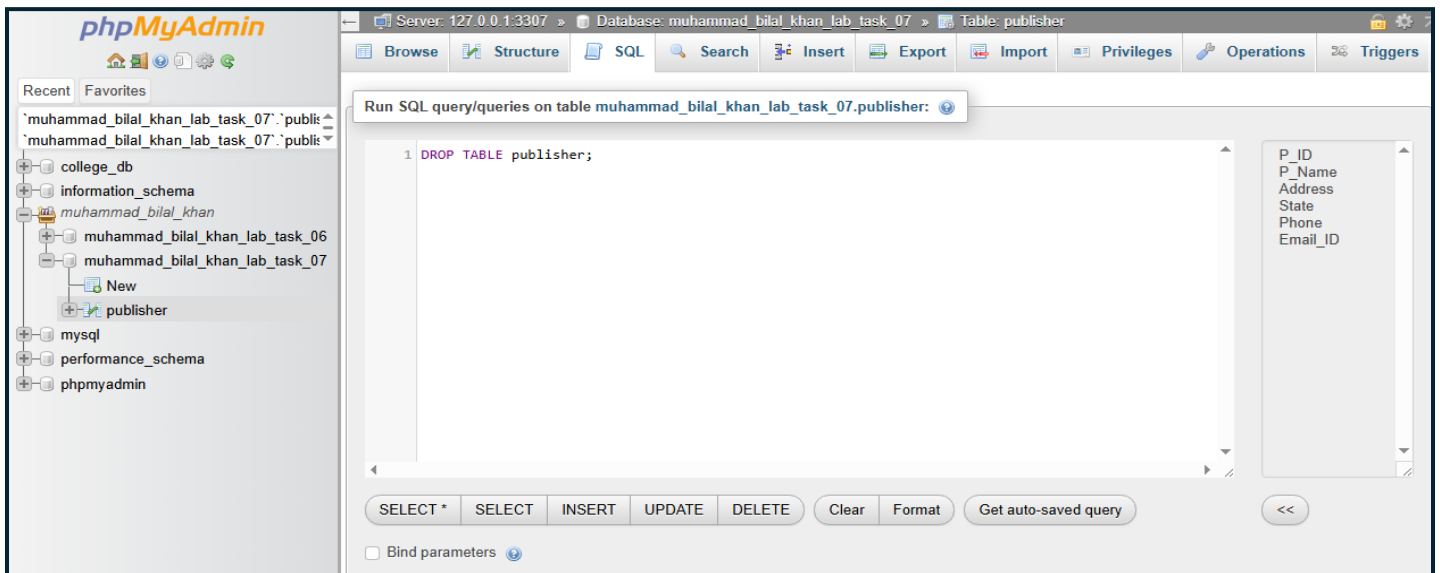
SELECT * SELECT INSERT UPDATE DELETE Clear Format Get auto-saved query

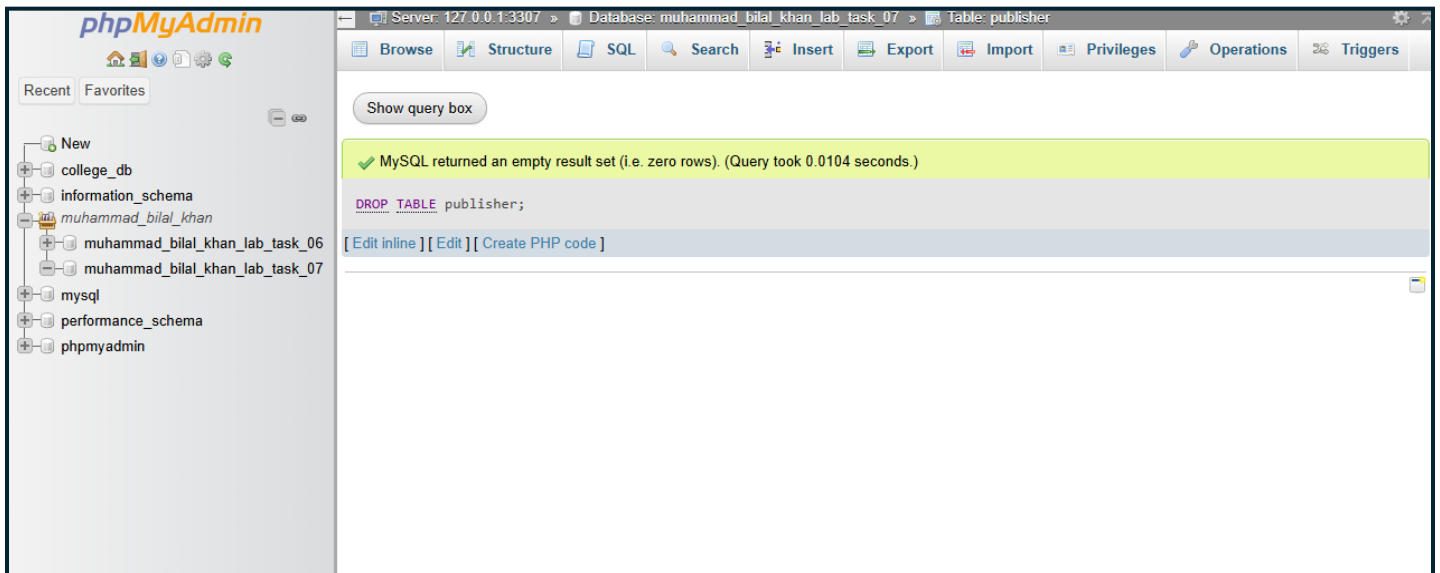
☐ Bind parameters

P_ID
 P_Name
 Address
 State
 Phone
 Email_ID



2. Delete the renamed table PUBLISHER using SQL.





Section 04:

Creating and Modifying Another Table Using SQL DDL Commands

1.Create table BOOKS with the following structure:

Attribute	Data Type	Field Size
ISBN	VARCHAR	4
Book_Title	VARCHAR	30
Category	VARCHAR	15
Price	NUMERIC	(6,2)
Copyright_Date	NUMERIC	4
Year	NUMERIC	4
P_ID	VARCHAR	4

phpMyAdmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_07

Structure SQL Search Query Export Import Operations Privileges

Run SQL query/queries on database muhammad_bilal_khan_lab_task_07:

```

1 CREATE TABLE BOOKS(
2 ISBN          VARCHAR(4),
3 Book_Title    VARCHAR(30),
4 Category      VARCHAR(15),
5 Price         NUMERIC(6,2),
6 Copyright_Date NUMERIC(4),
7 Year          NUMERIC(4),
8 P_ID          VARCHAR(4)
9 );

```

phpMyAdmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_07 » Table: books

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0014 seconds.)

[SELECT * FROM `books`](#)

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

ISBN	Book_Title	Category	Price	Copyright_Date	Year	P_ID
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Query results operations

[Create view](#)

2.Modify the BOOKS table using SQL DDL Commands:

Add a new column Page_Count with type NUMERIC(4).

phpMyAdmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_07 » Table: books

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

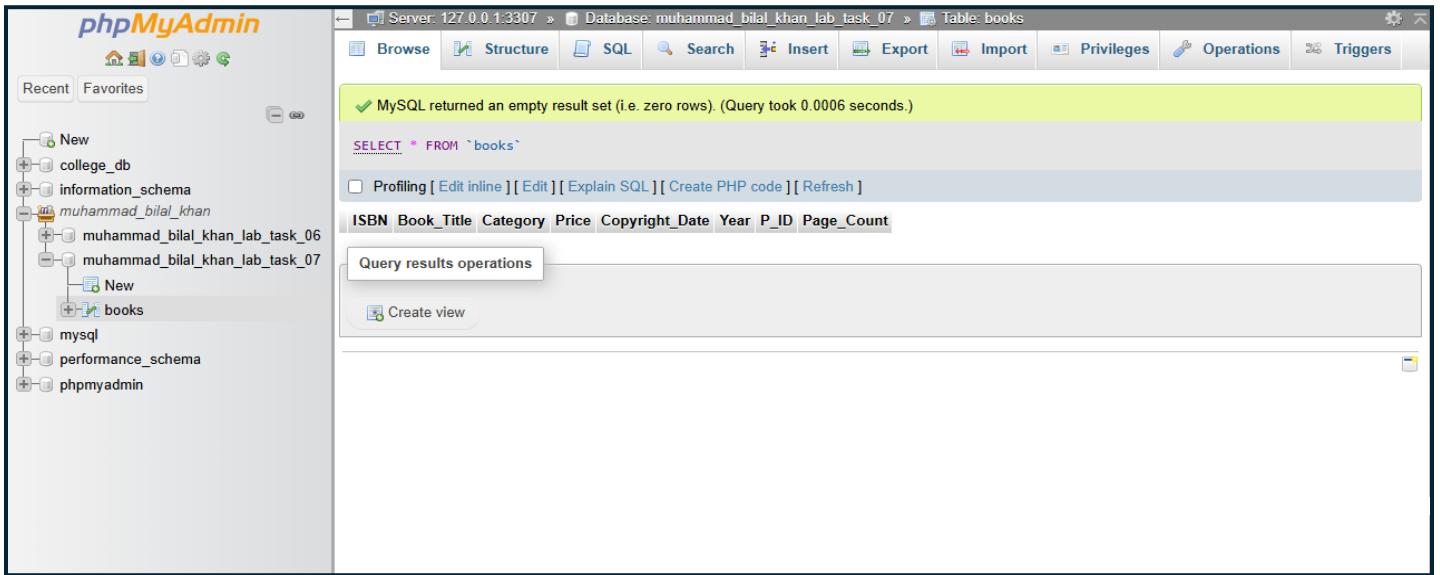
Run SQL query/queries on table muhammad_bilal_khan_lab_task_07.books:

```

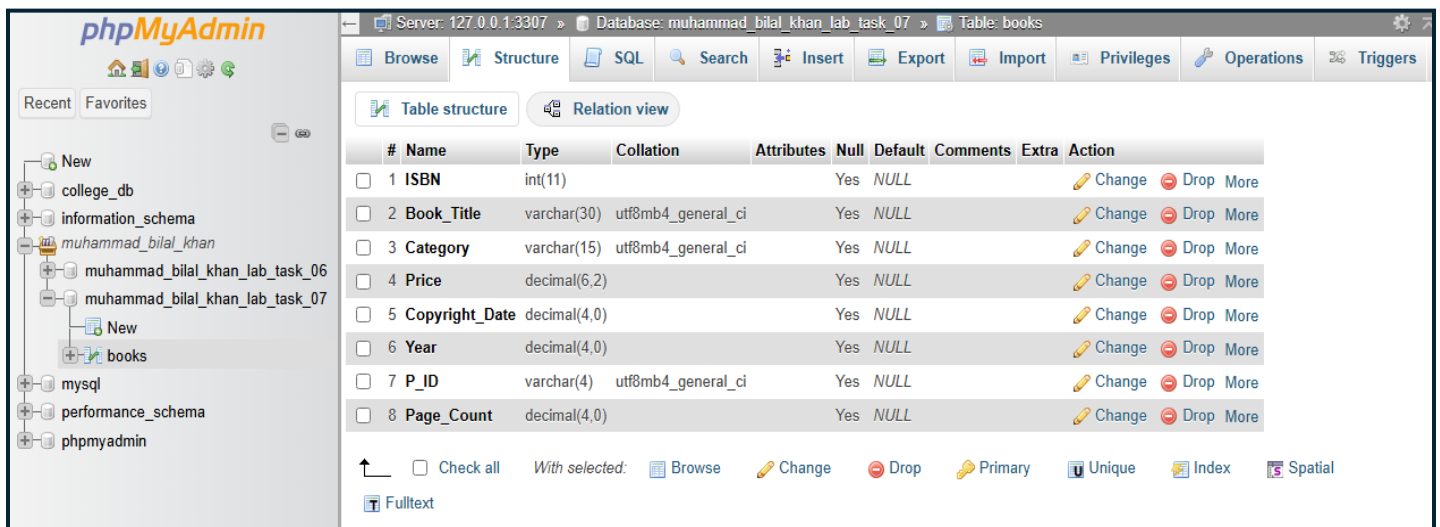
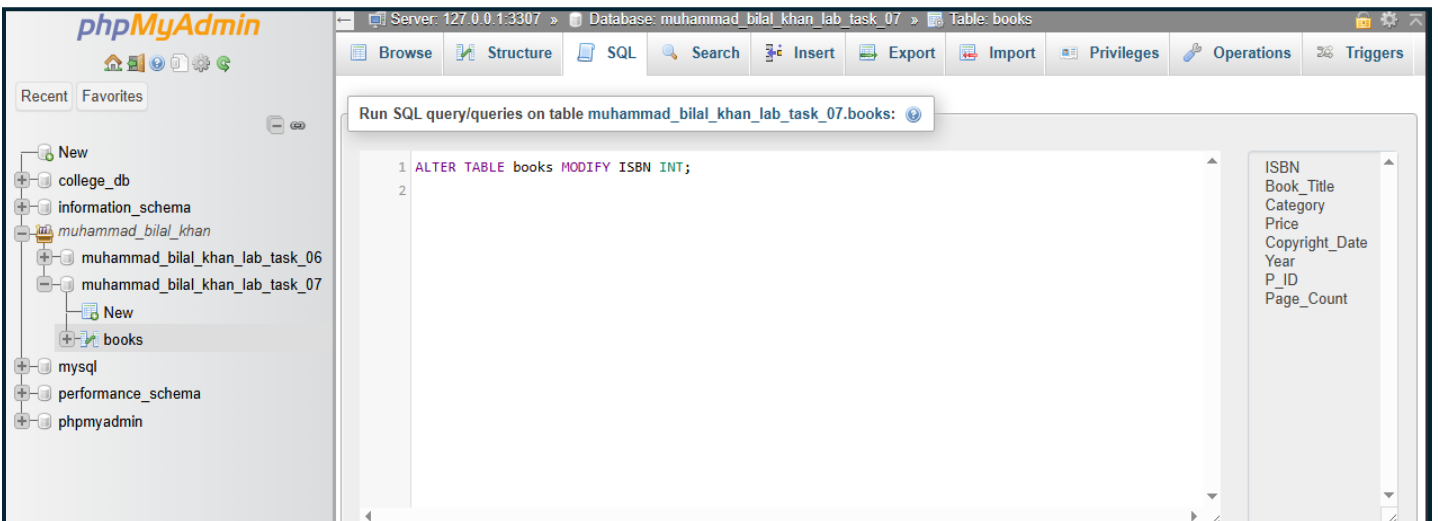
1 ALTER TABLE books ADD Page_Count NUMERIC(4);

```

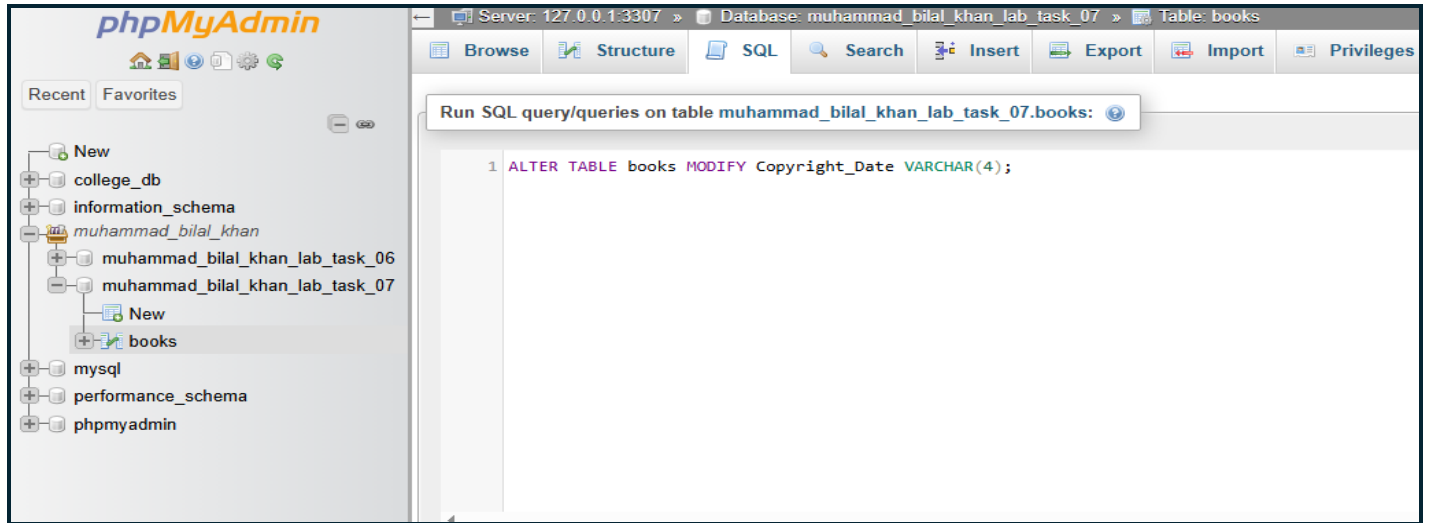
ISBN
Book_Title
Category
Price
Copyright_Date
Year
P_ID



Change the data type of ISBN from VARCHAR to INT.

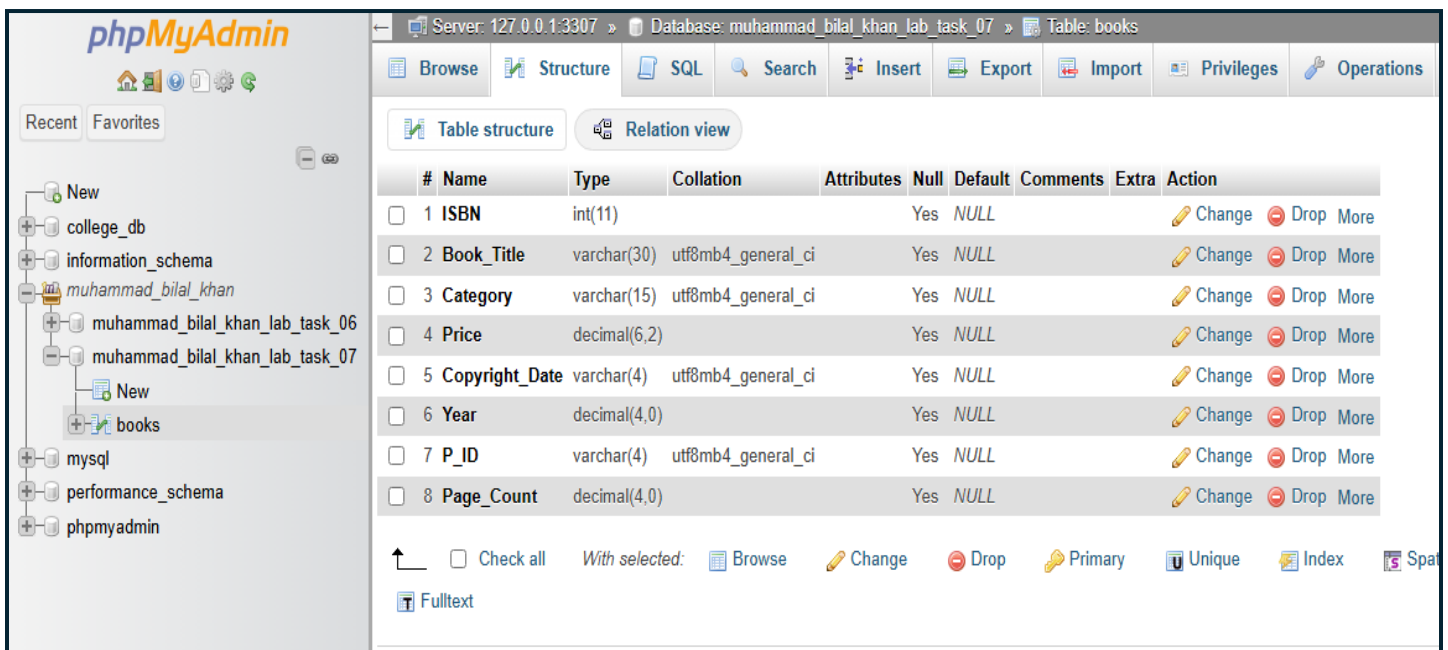


Change the data type of Copyright_Date from NUMERIC to VARCHAR.



The screenshot shows the phpMyAdmin interface. On the left, the database structure is visible, including 'college_db', 'information_schema', 'muhammad_bilal_khan', 'muhammad_bilal_khan_lab_task_06', 'muhammad_bilal_khan_lab_task_07', 'mysql', 'performance_schema', and 'phpmyadmin'. The 'books' table is selected under 'muhammad_bilal_khan_lab_task_07'. The main panel shows the 'SQL' tab with the following query:

```
1 ALTER TABLE books MODIFY Copyright_Date VARCHAR(4);
```



The screenshot shows the phpMyAdmin interface with the 'Table structure' tab selected. The table 'books' is selected under the 'muhammad_bilal_khan_lab_task_07' database. The table structure is displayed as follows:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 ISBN	int(11)			Yes	NULL			Change Drop More
☐	2 Book_Title	varchar(30)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	3 Category	varchar(15)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	4 Price	decimal(6,2)			Yes	NULL			Change Drop More
☐	5 Copyright_Date	varchar(4)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	6 Year	decimal(4,0)			Yes	NULL			Change Drop More
☐	7 P_ID	varchar(4)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	8 Page_Count	decimal(4,0)			Yes	NULL			Change Drop More

At the bottom, there are options to 'Check all', 'With selected:', 'Browse', 'Change', 'Drop', 'Primary', 'Unique', 'Index', and 'Spatial'.

Section 05:

Deleting Tables and Database

1. Use SQL DDL Commands to delete the BOOKS table.

phpMyAdmin

Recent Favorites

- New
 - college_db
 - information_schema
 - muhammad_bilal_khan
 - muhammad_bilal_khan_lab_task_06
 - muhammad_bilal_khan_lab_task_07
 - New
 - books
 - mysql
 - performance_schema
 - phpmyadmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_07 » Table: books

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on table muhammad_bilal_khan_lab_task_07.books:

```
1 DROP TABLE books;
```

ISBN
Book_Title
Category
Price
Copyright_Date
Year
P_ID
Page_Count

SELECT * SELECT INSERT UPDATE DELETE Clear Format Get auto-saved query

☐ Bind parameters

Apps voice maker Like4Like - Social M... Make QR f...

localhost says

Do you really want to execute "DROP TABLE books;?"

OK Cancel

phpMyAdmin

Recent Favorites

- New
 - college_db
 - information_schema
 - muhammad_bilal_khan
 - muhammad_bilal_khan_lab_task_06
 - muhammad_bilal_khan_lab_task_07
 - New
 - books
 - mysql
 - performance_schema
 - phpmyadmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_07 » Table: books

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/

```
1 DROP TABLE books;
```

ISBN
Book_Title
Category
Price
Copyright_Date
Year
P_ID
Page_Count

SELECT * SELECT INSERT UPDATE DELETE Clear Format Get auto-saved query

☐ Bind parameters

phpMyAdmin

Recent Favorites

- New
 - college_db
 - information_schema
 - muhammad_bilal_khan
 - muhammad_bilal_khan_lab_task_06
 - muhammad_bilal_khan_lab_task_07
 - mysql
 - performance_schema
 - phpmyadmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_07 » Table: books

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Show query box

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0087 seconds.)

DROP TABLE books;

[Edit inline] [Edit] [Create PHP code]

2. Delete the entire database from your system

(this can be done manually after taking a screenshot showing its deletion).

